

ACTIVITY -CSA0311

Stack : 29/11/26
 a) Push(54); Push(52); Pop(); Push(55); Push(62); S=Pop();
 (54, 52) (54) (54, 55) (54, 55, 62) (62)
 $S = 62$

enqueue(21); enqueue(24); dequeue(); enqueue(28); enqueue(32);
 $q = \text{dequeue}()$; (21) (24, 28) (24, 28, 32)
 (21, 24)

the (24) $q = 24$

$$S + q = 62 + 24 = 86$$

a) the result evaluating postfix expression 10 5+ 60 6/ * 8 - is

$$10 \ 5+ = 15$$

$$60 \ 6/ = 10$$

$$15 * 10 = 150$$

$$150 - 8 = 142$$

3) 8, 2, 3 ^ / 2. 3 * + 5 1 * - 1

Stack after first * evaluated;

Symbol	Stack
8	8
2	8, 2
3	8, 2, 3
^	8, 8, (2^3=8)
/	1 [8/8=1]
2	1, 2
3	1, 2, 3
*	1, 6 (2*3=6)

the top two elements are 1, 6

4. $a + b * * c - d ^ 1 e ^ 1 f$

character	stack	o/p
a	+	a
+	+	ab
b	+	ab
*	+	abc
*	+	abc*
c	-	abc*+d
-	-	abc*+d:
d	-	abc*+de
^	-	abc**+def
e	-	abc**+def
^	-	abc*+def
f	-	abc*+def
empty		

5. Stack operations:

	Stack	o/p
push(10)	10	-
push(10, 20)	10, 20	-
pop()	10	20
push(10)	10, 10	-
push(20)	10, 10, 20	-
pop	10, 10	20
pop	10	10
pop	empty	10
push(20)	20	-
pop	empty	20

Sequence of the popped values $\rightarrow 20, 20, 10, 10, 20$

